data format: 9600 8bits, 1 stop bit, no parity

orts modbus protocol 3 6 16, the data format is high bit first, low bit

message protocol format : read register, request frame

BYTE	0	1	2	3	4	5	6	7
finition of segm	ADDR	CMD	MSB	LSB	MSB	LSB	CRC-L	CRC-H
description	0xD2	0x03	ister starting addr		number of registers		CRC-16 checksum	

description : D2 03 00 0C 00 01 57 AA
D2=slave address, 03=read command, 000C=starting register address, 0001=number of registers, 57AA is the CRC-16 checksum

regular response frame

BYTE	0	1	2	3	4	5	6	7	8
definition	ADDR	CMD	Length	MSB	LSB	MSB	LSB	CRC-L	CRC-H
description	0xD2	0x03	L=2*N	value of 1st registe		value of n-th regi		CRC-16 checksum	

example:
D2 03 02 00 01 FC 56
D2=slave address, 03=read command, 02=length of data, 0001=reading the 1st register, the host command discharge status, FC56 is the CRC-16 checksum

message protocol format: write register, request frame

BYTE	0	1	2	3	4	5	6	7
definition	ADDR	CMD	MSB	LSB	MSB	LSB	CRC-L	CRC-H
description	0xD2	0x06	register address		data		CRC-16 checksum	

example : D2 06 00 0C 00 01 9B AA
D2=slave address, 06=write single holding register, 000C=register address, 0001=enable battery discharge, 9BAA is CRC-16 checksum

response

BYTE	0	1	2	3	4	5	6	7
definition	ADDR	CMD	MSB	LSB	MSB	LSB	CRC-L	CRC-H
description	0xD2	0x06	register address		data		CRC-16 checksum	

when writing a single register, the response frame has the same format as the request frame

message protocol format: write multiple registers, request frame

BYTE	0	1	2	3	4	5	6	7	8	9
definition	ADDR	CMD	MSB	LSB	MSB	LSB	MSB	LSB	CRC-L	CRC-H
description	0xD2	0x10	register address		register length (N)		data content (2N)		CRC-16 checksum	

example : D2 10 00 0C 00 02 00 01 00 02 7D 65
0xD2=slave address, 0x10=write multiple registers, 0x000C=starting address, 0x0002=write two consecutive registers, 0x0001=data for first register, 0x0002=data for second register, 0x7D65is the CRC-16 checksum

response

BYTE	0	1	2	3	4	5	6	7
definition	ADDR	CMD	MSB	LSB	MSB	LSB	CRC-L	CRC-H
description	0xD2	0x10	register starting address		register length (N)		CRC-16 checksum	

example : D2 10 00 0C 00 02 92 68
0xD2=slave address, 0x10=write multiple registers, 0x000C=starting address, 0x0002=write two consecutive registers, 0x9268 is the CRC-16 checksum

BMS protocol description						
data format: 9600 8 bits, 1 stop bit, no parity						
support modbus protocol,format is high bit first, low bit last						
address	type of data	R/W	content	range	data calculation instructions	
0x00~0x1F		R	cell voltages (two bytes per cell)	0--5000	0.001 3550*0.001=3.550v	
0x20~0x27		R	battery temperature	0--140 (-40 to 1	actual temperature = register value - 40 degrees C	
0x28		R	total battery voltage		0.1V 3507*0.1=35.0v	
0x29		R	current	0--65535	0.1A, 30000 Offset, discharge current = 0.1 * (register value - 30000)	
0x2A		R	state of charge	0--65535	0.001,800/1000=80%	
0x2B		R	max cell voltage	0--65535	1mv	
0x2C		R	min cell voltage	0--65535	1mv	
0x2D		R	max cell temperature	0--65535	1°C, temperature = register value - 40 deg C	
0x2E		R	min cell temperature	0--65535	1°C, temperature = register value - 40 deg C	
0x2F		R	charge/discharge status	0--65535	0=stationary, 1=charging, 2=discharging	
0x30		R	battery remaining capacity		0.1AH	
0x31		R	battery quantity			
0x32		R	number of temperature sensors			
0x33		R	battery cycle times			
0x34		R	Balance Stateequilibriam state	0--65535	0 : inactive 1 : active	
0x35		R	charge MOS status	0--65535	0=off 1=on	
0x36		R	discharge MOS status	0--65535	0=off 1=on	
0x37		R	average voltage	0--65535	1mV	
0x38		R	voltage difference	0--65535	1mV	
0x39		R	power	0--65535	W	
0x3A	unsigned int	R	fault status 1	0--65535	Byte 0 Bit 0: level 1 alarm - cell voltage too high Bit 1: level 2 alarm - cell voltage too high Bit 2: level 1 alarm - cell voltage too low Bit 3: level 2 alarm - cell voltage too low Bit 4: level 1 alarm - total voltage too high Bit 5: level 2 alarm - total voltage too high Bit 6: level 1 alarm - total voltage too low Bit 7: level 2 alarm - total voltage too low Byte 1 Bit 0: level 1 alarm - charging temperature too high Bit 1: level 2 alarm - charging temperature too high Bit 2: level 1 alarm - charging temperature too low Bit 3: level 2 alarm - charging temperature too low Bit 4: level 1 alarm - discharging temperature too high Bit 5: level 2 alarm - discharging temperature too high Bit 6: level 1 alarm - discharging temperature too low Bit 7: level 2 alarm - discharging temperature	
0x3B		R	fault status 2	0--65535	Byte 0 Bit 0: level 1 alarm - charging current too high Bit 1: level 2 alarm - charging current too high Bit 2: level 1 alarm - discharging current too low Bit 3: level 2 alarm - discharging current too low Bit 4: level 1 alarm - SOC too high Bit 5: level 2 alarm - SOC too high Bit 6: level 1 alarm - SOC too low Bit 7: level 2 alarm - SOC too low Byte 1 Bit 0: level 1 alarm - voltage difference too high Bit 1: level 2 alarm - voltage difference too high Bit 2: level 1 alarm - temperature difference too high Bit 3: level 2 alarm - temperature difference too high Bit 4~Bit7: Reserved	
0x3C		R	fault status 3	0--65535	Byte 0 Bit 0: charging MOS over-temperature warning Bit 1: discharging MOS over-temperature warning Bit 2: charging MOS temperature sensor failure Bit 3: discharging MOS temperature sensor failure Bit 4: charging MOS adhesion failure Bit 5: discharging MOS adhesion failure Bit 6: charging MOS circuit fault Bit 7: charging MOS circuit fault Byte 1 Bit 0: AFE acquisition chip failure Bit 1: single unit collection is offline Bit 2: single temperature sensor failure Bit 3: EEPROM storage failure Bit 4: RTC clock failure Bit 5: precharge failed Bit 6: vehicle communication failed Bit 7: internal network communication module failure	
0x3D		R	fault status 4	0--65535		
0x3E	unsigned int	R	remaining	0--65536	0.1km	
0x3F	unsigned int	R	reserved	0--65536	reserved	
0x40	unsigned int	R	reserved	0--65536	reseved	
0x41	unsigned int	R	reserved	0--65536	reseved	
0x42	unsigned int	R	reserved	0--65536	reseved	
0x43	unsigned int	R	reserved	0--65536	reseved	
0x44	unsigned int	R	reserved	0--65536	reseved	
0x45	unsigned int	R	reserved	0--65536	reseved	
0x46	unsigned int	R	reserved	0--65536	reseved	
0x47~0x56	unsigned int	R	single cell voltages (2 bytes per cell)	0--5000	0.001 3550*0.001=3.550v	
0x57			GPS valid flag	0--65535	valid : 0x45, other is invalid	
0x58			longitude	0--65535	0/W	
0x59~0x5A			longitude size	0--65535	0.000001	
0x5B			latitude	0--65535	S/N	
0x5C~0x5D			latitude size	0--65535	0.000001	
0x5E			speed	0--65535	0.1KM/H	
0x5F			altitude	0--65535	M	
0x60			number of satellites	0--65535	个	
0x61			protection board data acquisition normal	0--65536	valid : 0x45, other is invalid	
0x62~0x69			IMEI code	0--65537		
0x80		R/W	rated capacity	0--65535	0.1AH	
0x81		R/W	cell reference voltage	0--65535	mV	
0x82		R/W	number of acquisition boards	0--65535		
0x83		R/W	number of units in collection board 1	0--65535		
0x84		R/W	number of units in collection board 2	0--65535		
0x85		R/W	number of units in collection board 3	0--65535		
0x86		R/W	temperature for board 1	0--65535		
0x87		R/W	temperature for board 2	0--65535		
0x88		R/W	temperature for board 3	0--65535		
0x89		R/W	battery type	0--65535	0 : LiFePO4, 1 : ternary, 2 : lithium titanate	
0x8A		R/W	sleep wait time	0--65535	S	
0x8B		R/W	level 1 alarm - cell voltage too high	0--65535	mV	
0x8C		R/W	level 2 alarm - cell voltage too high	0--65535	mV	
0x8D		R/W	level 1 alarm - cell voltage too low	0--65535	mV	
0x8E		R/W	level 2 alarm - cell voltage too low	0--65535	mV	
0x8F		R/W	level 1 alarm - total voltage too high	0--65535	0.1V	
0x90		R/W	level 2 alarm - total voltage too high	0--65535	0.1V	
0x91		R/W	level 1 alarm - total voltage too low	0--65535	0.1V	
0x92		R/W	level 2 alarm - total voltage too low	0--65535	0.1V	
0x93		R/W	level 1 alarm - charging current too high	0--65535	0.1A, 30000 Offset	
0x94		R/W	level 2 alarm - charging current too high	0--65535	0.1A, 30000 Offset	
0x95		R/W	level 1 alarm - discharge current too high	0--65535	0.1A, 30000 Offset	
0x96		R/W	level 2 alarm - discharge current too high	0--65535	0.1A, 30000 Offset	
0x97		R/W	level 1 alarm - charging temperature too h	0--65535	1°C, 40 Offset	
0x98		R/W	level 2 alarm - charging temperature too	0--65535	1°C, 40 Offset	
0x99		R/W	level 1 alarm - charging temperature too	0--65535	1°C, 40 Offset	
0x9A		R/W	level 2 alarm - charging temperature too	0--65535	1°C, 40 Offset	
0x9B		R/W	level 1 alarm - discharge temperature too	0--65535	1°C, 40 Offset	
0x9C		R/W	level 2 alarm - discharge temperature too	0--65535	1°C, 40 Offset	
0x9D		R/W	level 1 alarm - discharge temperature too	0--65535	1°C, 40 Offset	
0x9E		R/W	level 2 alarm - discharge temperature too	0--65535	1°C, 40 Offset	
0x9F		R/W	level 1 alarm - excessive voltage differe	0--65535	mV	
0xA0		R/W	level 2 alarm - excessive voltage differe	0--65535	mV	
0xA1		R/W	level 1 alarm - excessive temperature dif	0--65535	1°C	
0xA2		R/W	level 2 alarm - excessive temperature dif	0--65535	1°C	
0xA3		R/W	balancing turn on voltage	0--65535	mV	
0xA4		R/W	equilibrium opening voltage difference	0--65535	mV	
0xA5		R/W	charging MOS switch		0=off or 1=on	
0xA6		R/W	discharge MOS switch		0=off or 1=on	
0xA7		W	SOC settings		0.001	
0xA8		R/W	MOS temperature protection alarm		1°C, temperature = register value - 40 deg C	
0xA9~0xB0		R	software version number	0--65535		
0xB1~0xB8		R	hardware version number	0--65535		
0xB9~0xC8		R	machine encoding			
0xC9~0xCB		R/W	parameter setting password			
0xCC~0xCD		R	production date	0--65535	year-month-date (YYMMDD00) in hex	
0xCE~0xE5		R	IP address	ASCII	if the address is 255.255.255.255:30096 the content sent is 0x32 35 35 35 2E 32 35 35 2E 32 35 35 2E 32 35 35 3A 52 30 30 39 36 20	followed by the port number
0xF0		W	system restart			
0xF1		W	system shutdown			
0xF2		W	return current to zero			
0xF3		W	reset			